RingMo-Sense: Remote Sensing Foundation Model for Spatiotemporal Prediction via Spatiotemporal Evolution Disentangling

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***Abstract*— Remote sensing (RS) spatiotemporal prediction aims to infer future trends from historical spatiotemporal data, e.g., videos and time-series images, which has a broad application prospect in many fields. The foundation model is a promising research direction for spatiotemporal information mining because of its robust feature extraction capability and has made rapid progress in natural scenes. Nevertheless, due to the spatially multiscale and temporally multiscale properties in RS data, these methods still encounter bottlenecks when applied to RS. Therefore, we propose a remote sensing foundation model for spatiotemporal prediction via spatiotemporal evolution disentangling, abbreviated as RingMo-Sense. Considering spatial affinity, temporal continuity, and spatiotemporal interaction, we construct spatial, temporal, and spatiotemporal triple-branch prediction networks. Specifically, we use parameter-sharing and progressive joint training strategies to achieve stable long-range prediction and parameter reduction simultaneously. In addition, we build an RS spatiotemporal dataset by collecting various RS videos and time-series images. The experimental results on six downstream spatiotemporal tasks demonstrate that the proposed model yields competitive performance.**

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I. Introduction

SPATIOTEMPORAL prediction refers to the mining of historical data for inherent patterns of change in order to deduce unknown trends and states. Humans have the ability to summarize data patterns to make future predictions through acquired learning, and models can also acquire this ability by learning from large amounts of data. In the field of remote sensing (RS), predictive models can contribute to a variety of application tasks, such as weather prediction [1], trajectory prediction [2], [3], and scene prediction [4]. In order to train a spatiotemporal predictive model with good performance, a large amount of spatiotemporal data is required, and the advancement of RS technology has led to a proliferation of RS data to meet this requirement. Some researchers have used these data to construct spatiotemporal datasets, such as mesoscale eddy tracking with satellite data [2], [3] and scene segmentation with aerial data [5]. Based on these data, some works have explored heuristic models for spatiotemporal prediction. However, these models are task-specific. There are many RS spatiotemporal prediction tasks; designing and training many task-specific models consume a tremendous amount of resources. Therefore, it is necessary to design a general spatiotemporal predictive model to meet the general needs of the RS field for a wide range of prediction tasks.

In the field of computer vision, the BERT-style pretraining approach has been proven to obtain general features and demonstrate outstanding performance on many image tasks. Now, there are some works [6], [7], [8], [9] to migrate the BERT-style training method to the foundation model of video recognition and achieve great results on downstream prediction tasks. BEVT [9] uses a two-branch network to decouple the model's learning of spatial and temporal information of data and achieves good results on image classification tasks. MaskViT [10] uses two window attention mechanisms and replaces the common fixed mask ratio with a variable mask ratio in the training strategy, thereby achieving excellent performance on video prediction tasks.

The existing foundation models are of great reference value but cannot be directly used for tasks in RS, for they are trained with few-object, simple-background natural scene videos [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18].

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Considering the vast differences between RS and natural scene data, these models need to work better when applied to RS prediction tasks by fine-tuning. These differences are shown in Fig. 1.

1) *Temporal Multiscale:* RS data include videos and time-series images, where time-series image data account for most of them. Time-series images are discrete in time, while the existing foundation models in computer vision can only deal with continuous videos, which are equidistant distributions of frames in the time dimension.
2) *Spatial Multiscale:* RS data have very complex backgrounds, and the sizes of different objects vary greatly. Foundation models in computer vision trained on data with clean backgrounds and centered objects impact the performance when applied to RS data.

For another challenge, existing predictive works do not explicitly model three characteristics of RS prediction simultaneously (see Fig. 2).

1) *Temporal Continuity:* The position of RS objects in discrete data changes continuously with time.
2) *Spatial Affinity:* In large-scale RS data, objects are far away but related to each other in categories.
3) *Spatiotemporal Interaction:* The interaction between multiple RS targets is dynamically evolving.

To alleviate the problems aforementioned, we follow the core idea of "spatial–temporal evolution disentangling" to design the RS foundation model for spatiotemporal prediction. In order to cope with the multiscale characteristics of RS data in time and space, the foundation model needs to handle temporal continuity and spatial affinity. Furthermore, the spatiotemporal evolution process of the dynamic interactions among multiple RS objects needs to be properly learned by the foundation model to realize the long-term stable prediction. Therefore, in this article, we propose a remote sensing foundation model for spatiotemporal prediction via spatiotemporal evolution disentangling (RingMo-Sense). Based on multiscale spatiotemporal RS data, we learn the spatial affinity, temporal continuity, and spatiotemporal interaction, respectively, by constructing spatial, temporal, and spatiotemporal three-branch predictive networks corresponding with blockwise, tubewise, and framewise masking approaches. On this basis, we adopt a progressive training strategy to realize the joint training of the three branches to share the parameter, achieving the stable prediction on RS data in complex scenes with large areas. To support the training of the predictive foundation model for RS, we built an RS spatiotemporal dataset by collecting multiple RS video data and time-series data. We use them for the pretraining of the foundation model and then fine-tune the pretrained model with the downstream dataset to obtain the model for specific tasks. By testing on multiple spatiotemporal prediction tasks (cloud forecasting, radar echo extrapolation, and RS video prediction) and spatiotemporal perception tasks (multiobject tracking, video detection, and time-series images' segmentation), our spatiotemporal predictive foundation model achieves competitive performance.

This article's contribution can be summarized into three parts.

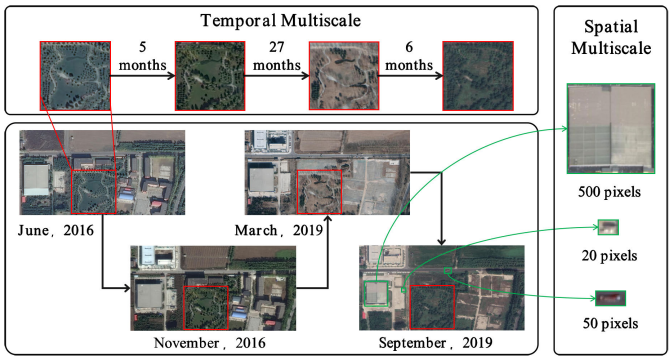


Fig. 1. Example of RS spatiotemporal data. In the temporal dimension, the intervals between different frames of RS data are unequal and irregular. In the spatial dimension, the background of RS data is complex, and the size of different objects varies greatly.

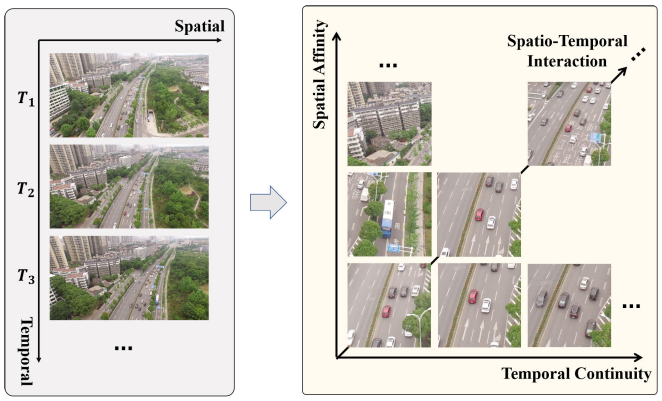


Fig. 2. Three characteristics of RS prediction, i.e., temporal continuity, spatial affinity, and spatiotemporal interaction.

1) We propose the first RS predictive foundation model (RingMo-Sense), which is pretrained on large-scale RS spatiotemporal data to obtain general predictive features and improve the performance of multiple downstream RS spatiotemporal tasks.
2) To tackle the multiscale problems in both spatial and temporal dimensions in RS data, we innovatively propose three-branch networks to model the spatial affinity, temporal continuity, and spatiotemporal interaction, respectively. In addition, a two-stage progressive joint training paradigm is designed to reduce parameters while enhancing the generalization performance.
3) Without any human supervision, we construct an RS spatiotemporal dataset. After deriving the predictive foundation model on the collected dataset, we fine-tune it on six typical RS tasks. The experiments show that our method achieves satisfactory result on six downstream datasets.

The rest of this article is organized as follows. In Section II, we give a brief introduction of the related works to this article. In Section III, we describe our approach in detail. Section IV contains a thorough description of our constructed dataset. Experimental results and an elaborate analysis are presented in Section V; Finally, in Section VI, we summarize our work and make some envisioning.

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II. Related Works

This section primarily introduces existing foundation models in RS, as well as spatiotemporal prediction in RS and spatiotemporal foundation models in computer vision.

A. Foundation Model in Remote Sensing

Foundation models use an auxiliary task (pretext) to construct supervision information from large-scale unlabeled data and train the network based on it. The foundation models learn the feature representation that is meaningful to downstream tasks in training. As the number of satellites has increased rapidly, massive RS images have been produced [19], [20], [21], [22], [23], [24]. In contrast to great success in self-supervised learning in natural scence [6], [10], [25], [26], researchers have become increasingly concerned with mining feature representations from massive unlabeled RS images.

Contrastive self-supervised learning. MoCo [27], [28], SimCLR [29], [30], and CMC [31] follow a classical contrastive learning approach.A number of data augmentation operations are manipulated to generate positive and negative samples. Inspired by CMC, in [32], SAR images, optoelectronic images, and label masks are used together to develop a multimodal representation for SAR semantic segmentation. In [33], unmanned aerial vehicle (UAV) recognition of rare wildlife is improved by combining contrastive learning methods with geometric enhancement.

Generative Self-Supervised Learning: Recently, the great success of the vision pretraining foundation models on the basis of Transformer architecture in natural scene vision tasks inspires the application of this architecture in the RS self-supervised pretraining foundation model. Sun et al. [34] propose RingMo, designing a foundation model training method for small and dense objects in complicated scenes. Wang et al. [35] adopt a Transformer architecture with about 100 million parameters. They propose rotating variable-size window attention to adapt to the large size of RS images and the arbitrary direction of the target. The model yields competitive performance with existing methods in detection, classification, and segmentation. Li et al. [36] propose a geographical knowledge (GeoKR)-driven representation learning and networks' pretraining method for RS images, which outperforms other self-supervised representation learning methods in terms of improving downstream task performance and reducing the demand for annotated data. Mañas et al. [37] propose seasonal contrast (SeCo), an effective pipeline to leverage unlabeled data for in-domain pretraining of RS representations. Models trained with SeCo achieve better performance than their ImageNet pretrained counterparts and state-of-the-art self-supervised learning methods on multiple downstream tasks.

B. Spatiotemporal Forecasting in Remote Sensing

At present, spatiotemporal forecasting in RS is mostly limited to meteorological prediction tasks. By mining the spatiotemporal dynamic characteristics of relevant meteorological elements in a certain range and a certain period, the model can understand the change law of the meteorological elements.

Partial research works predict precipitation over a short period utilizing extrapolation of radar echoes. In [38], [39], and [40], using linear extrapolation and correlation among adjacent echoes, researchers are able to solve smooth cases fast, but not complex echo motions. By simultaneously capturing transient changes and motion trends, Wu et al. [41] achieve better spatiotemporal evolution motion prediction. In [42], the hidden state transition is filtered to enhance the model's capacity to adapt to position change transformations such as rotation by applying a position change filter. Wang et al. [43] learn high-order spatiotemporal nonstationarity by differencing adjacent hidden states. There are also some works on weather prediction task based on satellite time-series images. Nielsen et al. [44] propose the CloudCast dataset based on weather satellites and a benchmark model to promote the research of satellite image time-series prediction. Dai et al. [45] propose a multiscale temporal conditional generative adversarial network to solve fuzzy prediction and accumulation of sequential error in the long-term forecast.

C. Sptiotemporal Foundation Model in Computer Vision

The huge success of image Transformers inspires the research of Transformer-based deep learning models for video recognition. In VTN [8], a video action recognition model with promising results is proposed, and the temporal attention encoder is added to the pretrained ViT model. In TimeSformer [7], factorized spatiotemporal attention leads to a good speed–accuracy tradeoff. It is shown empirically by Liu et al. [6] that Video Swin Transformer with spatiotemporal locality bias outperforms other vision Transformers. BEVT [9] decouples video representation learning into spatial information representation learning and temporal dynamic information representation learning. When training with real-world scenes, MaskVit [10] uses two types of attention, spatial and spatiotemporal, and uses varying mask ratios.

III. Method

We propose the foundation model (RingMo-Sense) for spatiotemporal prediction tasks, which is specially designed for the field of RS, and the architecture is shown in Fig. 3. In order to obtain the modeling ability of RS spatiotemporal data and reduce training difficulty, the model adopts a multibranch network structure and a progressive training strategy. This model was pretrained in a self-supervised training method for multimask reconstruction on massive RS data. Next, we will elaborate on the details on the design of RingMo-Sense and the training strategy.

A. Preliminary

Suppose that we have constructed a large unlabeled RS dataset $\mathcal{D} = (x_1, x_2, \ldots, x_n)$, where x_i represents a sample of input, such as an RS video clip. The goal of pretraining is to learn an encoder model f_θ in an unsupervised manner, where θ represents the parameters of the encoder, enabling the model to encode the input samples and generate meaningful representations.

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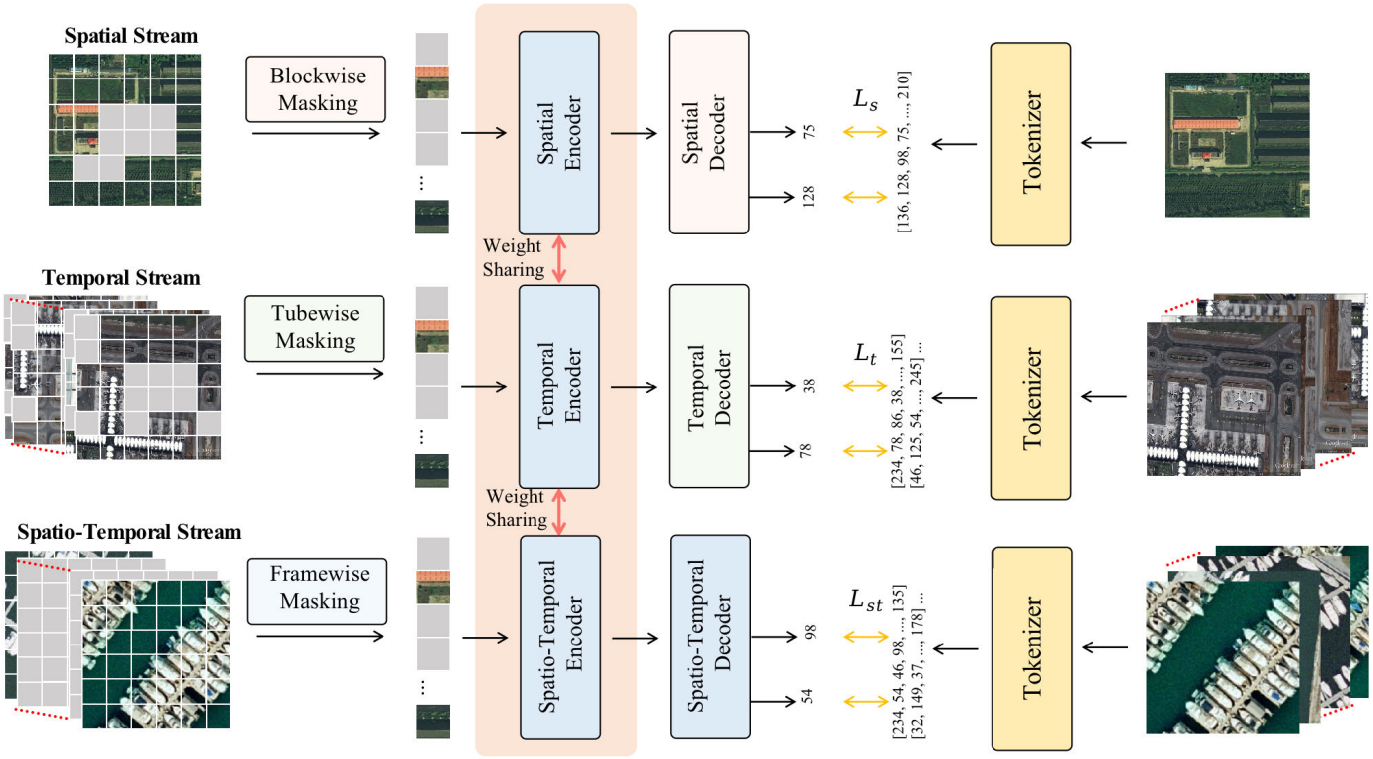


Fig. 3. Architecture of spatiotemporal predictive foundation model (RingMo-Sense) in the field of RS. The data will be input into the model and compared with the tokens generated by the tokenizer. We have three data streams here, and we first train the first two streams and then train the three streams together. The encoder is the Video Swin Transformer.

Specifically, the input samples [Input] of a minibatch are first preprocessed and precoded. After being masked randomly, the [Input_M] is obtained and sent into the encoder f_θ. Then, the expressive features $f_\theta(\text{Input}_M)$ are decoded and produce the predicted results V_P to compare with input samples. The target of pretraining is to minimize the gap between label $\mathcal{T}$(label) and output V_P. Here, the tokenizer $\mathcal{T}(\cdot)$ is utilized to compress the ground truth and improve training efficiency. After fine-tuning specific RS datasets through supervised methods, the requirements of downstream tasks can be realized.

B. Overall Pipeline of RingMo-Sense

We design the model as a three-branch structure and implement three disentangled training branches (i.e., the spatial branch, the temporal branch, and the spatial–temporal branch) by different masking strategies so that different branches learn different modeling capabilities. In addition, the weight-sharing strategy is applied to the weight parameters of the three branches in order to reduce the number of parameters in the model. In Section III-B, we first introduce the core architecture of the RingMo-Sense and then provide the details supporting this architecture, including the generation of patches, the masking strategy, the weight-sharing strategy, and the objective function.

1) Disentangled Learning Architecture: There are three branches in our model, and the foundation model of all branches is based on the Video Swin Transformer [6]. The details of each branch are shown in Fig. 4, where T equals 1 for the spatial branch but greater than 1 for the other two branches.

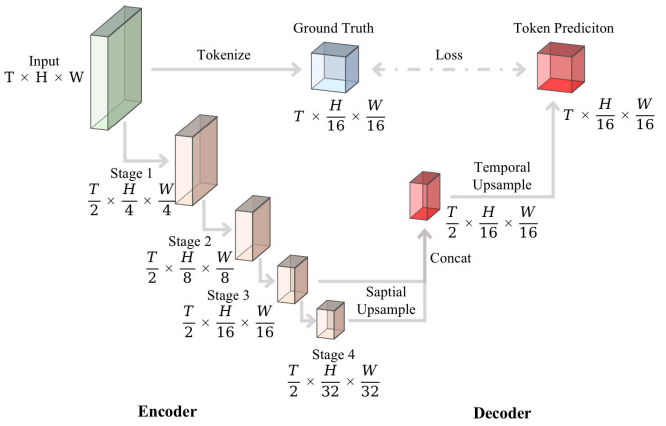


Fig. 4. Encoder and decoder of our foundation model. The input is tokenized to acquire the ground truth, and an encoder–decoder foundation model transforms the input to the prediction and calculates the loss between the prediction and the ground truth. The encoder in the figure is a Video Swin Transformer with four stages. The decoder is composed of simple transposed convolution layers and linear layers.

The encoder is composed of a spatial branch, a temporal branch, and a spatiotemporal branch. The spatial branch is dedicated to learning the spatial relative positional relationship between objects in an image and solving spatial multiscale problems. The temporal branch models the continuous change process of the object position in time-series RS images or videos to solve temporal multiscale problems. Furthermore, the spatiotemporal branch is designed to model the dynamic interaction between multiple targets and improve the performance of long-term stable prediction.

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The decoder is at the right part of the Fig. 4. It consists of the following three parts: a transposed convolution layer for upsampling the features spatially from Stage 4, a linear layer that is designed for fusing the concatenated features derived from Stage 3 and the last upsample stage, and another transposed convolution layer for upsampling the features temporally.

For the input image or video data, after implementing different linear layers and masking strategies, informative features with reduced spatial resolution and an expanded number of channels can be obtained using the Video Swin Transformer encoder. Then, the features are fed to different upsampling layers, followed by a shared lightweight decoder set to compute the final CrossEntropy loss. The aforementioned process can be formulated as

$$\begin{aligned} f_s &= F_{2d}\left(\text{Input}_s\right) \\ f_t &= F_{3d}\left(\text{Input}_t\right) \\ f_{st} &= F_{3d}\left(\text{Input}_{st}\right) \\ \text{output}_* &= \text{LinearDecoder}(f_*) \end{aligned} \tag{1}$$

where Input_s, Input_t, and Input_{st} denote the inputs of the spatial encoder, the temporal encoder, and the spatiotemporal encoder, respectively, and after the encoding process through 2-D or 3-D convolution, the features f_s, f_t, and f_{st} are obtained. The final output of the model is obtained by adding a linear layer to the features mentioned above, where $*$ denotes s, t, or st.

2) Generation of Patches: For the spatial branch, during training, the input image $\text{Input}_{\text{img}} \in \mathbb{R}^{H\times W\times 3}$ is transformed into multiple $4 \times 4 \times C$ patches and then multiplied with a mask, and the pixel values at the masked locations are initialized randomly. For the temporal and spatiotemporal branches, the frames of the input video are set to T, and then, the input $\text{Input}_{\text{video}} \in \mathbb{R}^{T\times H\times W\times 3}$ is converted into $2 \times 4 \times 4 \times C$ 3-D patches. In addition, in the specific practice, to be consistent with the video format and train simultaneously, we splice the time-series RS images into video as the input of the temporal branch and the spatiotemporal branch.

3) Mask Strategy: As shown in Fig. 5, to allow specific branches to obtain designated modeling capabilities and help model learn different representations, we design three masking strategies and employ them in different branches. For the spatial branch, we utilize the blockwise masking strategy to help focus on the spatial features; given the range of the mask proportion, the position and aspect ratio of the mask are randomly selected within a certain range. For the temporal branch, a tubewise masking strategy is used to help capture the temporal information, which is directly expanded from the blockwise masking strategy in the time dimension, i.e., the starting position and the length of the tubewise mask in the time dimension are randomly determined within a certain range, and then, a mask generated according to the blockwise masking strategy is executed for all determining video frames. Note that the single-frame mask in the temporal branch and the mask in the spatial branch are generated under the same strategy but may be different due to the randomness of some settings. For the spatiotemporal branch, a framewise masking strategy is used to help understand spatial–temporal dynamic interactions, where the temporal start position of the mask is randomly determined within a certain range for the input slices, and the length of the mask is constant; from the start position, all pixels of the frames within the length of the mask are masked. Thus, different mask strategies for the three branches allow the model to obtain different modeling capabilities, i.e., spatial, temporal, and spatiotemporal modeling capabilities simultaneously.

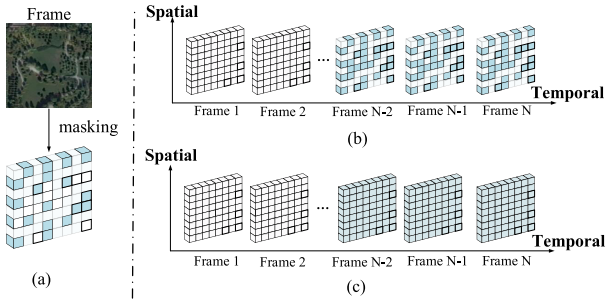


Fig. 5. (a) Blockwise, (b) tubewise, and (c) framewise masking strategies. Blue means masking, while gray means raw pixels.

4) Weight Sharing Strategy: The main structures in Transformer, such as the attention module and the MLP layer, are concerned with the irregular dimensionality of the input token, and before being input to different branches separately, the image and video tokens are passed through different linear layers to ensure that the dimensionality of the tokens is consistent. Moreover, the mask operation obviously does not change the dimensionality of the token. For these reasons, we have the following weight-sharing strategy: the parameters of the temporal and spatiotemporal branches are completely shared, and most of the parameters of the spatial branch are shared with the other two branches, except for operations related to data types, such as upsampling layers. While reducing the number of parameters in the model, weight sharing also allows the model to learn multiple modeling capabilities with the same parameters.

5) Objective Function: We adopt the CrossEntropy loss as the objective function to train our model, which can be defined as follows:

$$\begin{aligned} L_s &= \text{CrossEntropy}\left(\text{output}_s, \mathcal{T}(\text{label}_s)\right) \\ L_t &= \text{CrossEntropy}\left(\text{output}_t, \mathcal{T}(\text{label}_t)\right) \\ L_{st} &= \text{CrossEntropy}\left(\text{output}_{st}, \mathcal{T}(\text{label}_{st})\right) \\ L &= L_s + L_t + L_{st} \end{aligned} \tag{2}$$

where $\mathcal{T}(\text{label}_s)$, $\mathcal{T}(\text{label}_t)$, and $\mathcal{T}(\text{label}_{st})$ are ground-truth tokens obtained by applying a tokenizer on respective inputs. $\mathcal{T}(\cdot)$ represents the tokenizer, and L denotes the total loss, which is the sum of the losses incurred by the three branches.

C. Training Strategies

To alleviate the difficulty of model training caused by multiple branches, we distribute the pretraining of the model into two training phases, and the progressive training strategy reduces the difficulty of training. The model also goes through

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a fine-tuning step before being applied to a specific downstream task. In this section, the pretraining and fine-tuning methods are described in detail.

1) Progressive Pretraining Method: It is challenging for the spatiotemporal foundation model to obtain general abilities simultaneously since different branches focus on various information. For this reason, we use a two-stage progressive training strategy to allow the model to gradually learn higher dimension modeling capabilities, which reduces the difficulty of training and improves effectiveness and efficiency. In the first stage, with parameters initialized randomly, we train the spatial branch and the temporal branch using image data and video data, allowing the model to learn the spatial and temporal information of the data separately. After the first-stage pretraining, the spatiotemporal branch is added to the training and is initialized with the pretrained temporal branch parameters obtained from the first stage. Based on the first stage of training, with the help of the blockwise masking strategy and the tubewise masking strategy, the model has a certain modeling ability for spatial and temporal information. Since the masking strategy of the spatiotemporal branch is to mask all the selected video frames, the difficulty for the model to recover the masked pixels is greatly increased after adding the spatiotemporal branch, which will strengthen the model's ability to model spatiotemporal information. The detailed progressive training process is shown in Algorithm 1.

Algorithm 1 Progressive Pretraining

```
stage = 1
while stage ≤ 2 do
  if stage is 1 then
    train(spatial, temporal)
    weight initialized randomly
    f_s = F_2d(Input_s)
    f_t = F_3d(Input_t)
    L = L_s + L_t
    weight saved as W_1
  else {stage is 2}
    train(spatial, temporal, spatiotemporal)
    weight initialized with W_1
    f_s = F_2d(Input_s)
    f_t = F_3d(Input_t)
    f_s−t = F_3d(Input_st)
    L = L_s + L_t + L_st
    weight saved as W_final
  end if
end while
```

2) Downstream Task Fine-Tuning: To accommodate multiple types of downstream tasks, the model is applied to the downstream tasks by discarding the lightweight decoders used in pretraining and replacing them with the decoders specific to the downstream tasks. Based on the parameters obtained from the pretraining, RingMo-Sense can be fine-tuned with data from the downstream tasks.

IV. Dataset

A. Dataset Collection

The design of the RS foundation model for spatiotemporal prediction aims to provide a generalized framework. This framework is designed to generalize each spatiotemporal-related task of RS. Also, it can adapt to the task requirements and enhance the robustness and generalization capability of different tasks. After determining the fundamental direction of data collection (i.e., spatiotemporal data related to RS/aerial-space scenes), we attempt to explore as much as possible the multisource property of data, the diversity of scenes, and the multiscale property of platforms. In summary, the principle of collecting datasets for the RingMo-Sense generic pretraining model is to enhance the breadth of data coverage, enrich the depth of data content, and ensure the robustness of data distribution under the proposition of spatiotemporal-related datasets. Guided by the aforementioned principles, this dataset can provide a stable and generic pretraining model for spatiotemporal associated evolutionary tasks in the context of RS.

The pretraining dataset is processed and constructed according to the current mature spatiotemporal data processing method. The principle of data processing is to maintain the original distribution of the data as much as possible. Simultaneously, we aim to preserve the original spatiotemporal information of the data while enhancing its temporal and spatial dimensions to the fullest extent possible. Then, this dataset can serve as the data basis in support of the robustness of the model. The specific data processing and dataset construction pipeline are given as follows.

1) Process the format of the collected datasets, and unify the different formats (mp4, avi, temporal images, and so on) into avi video format to facilitate a uniform reading interface for the model.
2) To crop the video data in the spatial dimension, we employ a sliding window approach with a fixed size of 448 × 448 and a 50% overlap rate along the width and height dimensions.
3) In the temporal dimension, we use the strategy of frame skipping and overlap cropping, where original time-series data are extracted at intervals to form two subseries. Then, based on the subseries, we crop the video into 84 frames, with 14 frames overlapping between adjacent videos.

After the above simple processing, we obtained 1 025 000 training data and verified by sampling that the dataset distribution did not shift dramatically from the previous one.

B. Dataset Description

Based on the dataset selection principles proposed in Section IV-A, we collect a dataset for the RingMo-Sense model. In general, this dataset is characterized by multiformat, multiplatform, and multiscenarios, which will be described in Section IV-B1–IV-B3.

1) Data-Related Statistical Information: A total of 15 000 raw data are collected, after retrieval and screening, as shown in Table I. To better fit the model training and to ensure

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TABLE I
STATISTICAL INFORMATION OF DATASETS FOR SPATIOTEMPORAL RS FOUNDATION MODEL TRAINING

Dataset	Platform	Number	Duration /min	Resolution	Categories
Okutama-Action [46]	UAV	43	43	3840×2160	12
AU-AIR [47]	UAV	32,823 frames	120	1920×1080	8
Stanford Drone Dataset [48]	UAV	60	-	1424×1088	8
ERA [49]	UAV	2864	239	640×640	25
UAV123 [50]	UAV	123	63.5	1280×720	9
VISO [51]	Satellite	40	-	12000 × 5000	4
UCLA Aerial Event Dataset [52]	UAV	27	-	2386×1723	12
Drone-tracking Dataset [53]	UAV	70	8.8	1280×720	-
UAVDT [54]	UAV	100	20.6	1080×540	3
WebUAV-3M [55]	UAV	4500	1734	-	223
SatSOT [56]	Satellite	105	-	12000×5000	4
Small Aircraft Target Detection [57]	UAV	22	-	256×256	-
SEA DRONES SEE [58]	UAV	22	-	3840×2160	-
Self-built data	Satellite	203	-	1920×1080	2

that the data distribution is not corrupted, only a simple spatial scale crop and a common temporal frame extraction are performed on the data according to the pipeline described in IV-A. After such processing, roughly 1 025 000 data are obtained, which are sufficient to support adequate training of the RingMo-Sense foundation model. In addition, our data are gathered from diverse sources and encompass a wide range of scenarios, thus enabling the model to learn more versatile and resilient generic features.

2) Data Acquisition Platforms: Our data are mainly derived from two types of RS platforms, UAV, and satellite-based sensors. The data obtained from different types of platforms with distinct parameters have very diverse properties. The data acquired by the UAV platform have relatively higher resolution, richer scenes, and higher content granularity. Due to its platform characteristics, the acquired data generally have a certain tilt angle, which can provide multiple perspectives for the perception of content objects. With its platform characteristics, the data acquired by the satellite-based platform have relatively lower granularity, but its coverage area is larger, the richness of content categories is more plentiful, and the distribution levels within the same scene data are also more abundant. These two types of platforms have more diverse data variations caused by parameter settings and shooting conditions (lighting, shading, clouds, fog, and so on). The rich variation and differences between and within datasets force the model to learn more essential spatiotemporal correlation features, making our model perform well on all kinds of downstream tasks.

3) Scene Richness: The richness of the scenes determines, to a certain degree, the model's ability to generalize across various application conditions. We have taken it into account when collecting data, so we intentionally enhance the kinds of scenes covered by our dataset. The scene scales covered by our dataset range from satellite-based low-resolution low-dynamic urban building complexes to airborne high-resolution high-dynamic human activities. Almost all scenes in the currently available datasets in the domain are covered. Such scene richness allows the model to learn richer and more diverse base features, relieves the pressure of fine-tuning different scene tasks downstream, and improves the performance of grounded applications.

V. EXPERIMENTS

A. Foundation Model

The main structure of the model is Video Swin-Base, the optimizer is AdamW [59], the initial learning rate is set to 0.0005, and the decay strategy is cosine annealing. Experiments are implemented by the MindSpore framework. For the first stage of pretraining, the batch sizes are set as 64 and 8 for the spatial branch and the temporal branch, respectively, and the slice length of the input videos is 16. For the second stage of pretraining, the batch size of the spatial branch is 32, and the batch size of both temporal and spatiotemporal branches is 4. The size of the images and videos is 224 × 224. As shown in Algorithm 1, we initialize the parameters of RingMo-Sense randomly for the first pretraining stage, and in the second pretraining stage, we initialize the parameters of RingMo-Sense with parameters obtained in the first pretraining stage. For all downstream tasks, we initialize the parameters of RingMo-Sense with parameters obtained in the second pretraining stage. The comparison methods such as YOLOX are the current state-of-the-art methods in downstream tasks. For fair comparisons on each specific downstream task, we fine-tune the comparison methods and the proposed RingMo-Sense on the same RS training set and obtain the experimental score on the testing set. For RingMo-Sense, the tokenizer is selected as DALL-E [60]. For blockwise masking strategy, we mask 50% of the patches, and the position of the mask is randomly chosen; for tubewise masking strategy, we first generate a mask in the same way as the previous strategy and then replicate it to generate a tube mask, where the length of the tube mask is determined within the range of $[0.5, 1]$ of the whole frames and the starting position is determined randomly; for framewise masking strategy, the starting position

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is randomly chosen, and the mask ratio is 50%. To verify the contribution of each component in the proposed RingMo-Sense framework, we conduct abundant ablation studies, and the relevant experimental settings are elaborated on in the following. In this article, "RingMo-Sense" denotes the whole foundation model with the progressive pretraining paradigm of spatial (S) stream, temporal stream (T), and spatiotemporal (O) stream if there is no special note. "RingMo-Sense(-J)" denotes the foundation model with the three-stream joint pretraining strategy, and it is used to justify the effectiveness of the proposed progressive pretraining paradigm. To demonstrate the contribution of the spatial stream, we obtain the model weights, i.e., RingMo-Sense(-TO), with the pretraining of temporal and spatial–temporal streams. Similarly, the pretrained weight RingMo-Sense(-SO) is obtained to investigate the effect of the temporal stream, and RingMo-Sense(-ST) is used to explore the effect of the spatial–temporal stream. Furthermore, to prove the effectiveness of the proposed spatial–temporal masking method, RingMo-Sense(-STT) is obtained by employing the same tubewise masking method as the temporal stream in the spatial–temporal stream.

B. Video Prediction in Remote Sensing

Human beings have a strong ability to predict future changes in things. Based on the transcendental understanding of the objective development laws of things and the perception of the current environment, people can infer and predict their future trends. For example, based on past driving experience and visual perception of current road conditions, drivers can determine the trajectory of pedestrians moving ahead and, thus, avoid traffic accidents. Recent research [61], [62] in computer vision has gradually begun to pay attention to humans' predictive coding capabilities.

By analyzing a video clip's history frames, one can predict the future frames and an internal representation model that can accurately model the content and dynamics of the video; this is why video prediction is considered as a promising research direction for unsupervised representation learning. Video prediction in RS aims to capture the spatial features and temporal dynamics in RS videos, specifically in terms of changing patterns, such as the interaction of objects in the scene and the shift of the background. This is a very interesting element of research with important implications for tasks such as natural disaster early warning.

1) Specific Architecture: As shown in Fig. 6, the model architecture for RS video prediction is an adaptation of the classical model SimVP [63] for natural scene video prediction. We replace the backbone of SimVP with the Video Swin Transformer used in our foundation model, which consists of four stages in a cascade. The spatiotemporal features are then learned by the SimVP Translator, and finally, by deconvolution decoding, the feature map is restored to its original size.

2) Dataset Introduction: MOR-UAV [64] is a massive video dataset for moving object recognition (MOR). It contains 30 UAV videos, annotated with 89 783 moving object instances (e.g., car and heavy vehicle), consisting of 10 948 frames. It involves different scenarios such as weather conditions,

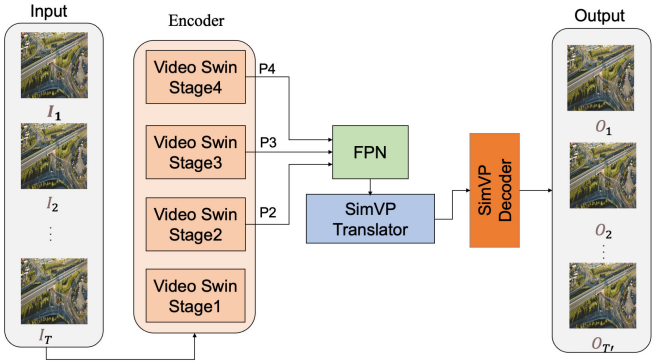


Fig. 6. Architecture of the video prediction in RS with the foundation model. The translator and the decoder are built on SimVP, adopting the spatiotemporal encoder of the foundation model as the backbone. I_1–I_T represent the input frames, and O_1–$O_{T'}$ denotes the output frames.

TABLE II
ABLATION STUDIES ON VIDEO PREDICTION

Model	MSE ↓	MAE↓	SSIM↑
PredRNN[66]	3206.53	18731.13	0.5301
RingMo-Sense(-J)	1775.68	11334.10	0.6905
RingMo-Sense(-TO)	2042.44	13144.97	0.6704
RingMo-Sense(-SO)	1819.16	11362.44	0.6876
RingMo-Sense(-ST)	1994.15	12516.83	0.6665
RingMo-Sense(-STT)	1894.04	11522.40	0.6770
RingMo-Sense	1766.97	11083.61	0.6955

occlusions, flight height changes, and multiple camera views. In this article, we use only the video data in the dataset and do not use any other information.

3) Implementation Details: In the experiments, we take 20 continuous frames from the video as one sample, with the first ten as input and the rest as labels, all of which are resized to a size of 256 × 256. We utilize the Adam optimizer [65] by default, the OneCycleLR learning rate strategy with a default learning rate of 0.01 and a batch size of 16, and trained 100 epochs. We used mean squared error (MSE), mean absolute error (MAE), and structural similarity index (SSIM) criteria to evaluate the prediction performance. SSIM is taken as a similarity measure between two given images and is given as follows:

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{3}$$

where μ_x and μ_y denote the average values of x and y, respectively. σ_x^2 and σ_y^2 denote the variance of x and y, respectively. C_1 and C_2 are all positive constants.

4) Comparisons With SOTA Methods: To demonstrate the effectiveness of our model with RingMo-Sense parameters, we compared the results of our model with PredRNN [66] for video prediction in natural scenes. When we use the baseline model PredRNN [66], which outperforms many existing models in natural scenes video prediction, to predict future frames of UAV videos, the SSIM could achieve 0.5301. Loading RingMo-Sense's pretraining parameters, the SSIM reached 0.6955, which performed better than PredRNN.

5) Ablation Studies: To verify the effectiveness of each component of RingMo-Sense on RS video prediction task,

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we load the corresponding pretraining parameters according to the ablation experimental configuration described in Section V-A, and the results are given in Table II. MSE of the RingMo-Sense is 1766.97, which is 8.71 lower than that of RingMo-Sense(-J), proving the effectiveness of our proposed progressive training paradigm; it is 275.47 and 52.19 lower than RingMo-Sense(-TO) and RingMo-Sense(-SO), respectively, proving the effectiveness of the spatial and temporal branches, respectively; MSE reaches 1994.15 under the RingMo-Sense(-ST) configuration, indicating that the spatiotemporal branch plays a positive role in the RS video prediction task. With RingMo-Sense-(STT) configuration, the MSE is 1894.04, which is 127.07 higher than that of RingMo-Sense, proving that our proposed spatiotemporal mask strategy is superior to the tubewise strategy for extracting spatiotemporal features.

6) Visualization Results: The visualization results of the RS video prediction task are shown in Fig. 7. Although PredRNN can extract spatiotemporal features, the prediction results are very vague, which is disastrous for RS scenes where the object scale is already small. Comparatively, the prediction results of the RingMo-Sense model are much clearer and closer to the ground truth. Fig. 8 shows the mean value of the predicted results (statistical average of the gray scale of each image frame) for a random sample in the MOR-UAV dataset using our method and PredRNN, which demonstrates that our predicted results are much closer to the ground truth. The downside is that our model is still deficient in capturing motion patterns, and further analysis of motion patterns may be introduced in the future with interframe differential information.

C. Cloud Forecasting

The derivation of cloud formation and development plays a very important role in modern weather forecasting systems. Inaccurate cloud forecasts may affect the accuracy of the overall weather forecast. Cloud forecasting is an important challenge that remains unsolved in meteorology, and there is also little research addressing it from the perspective of machine learning.

In this article, cloud forecasting is categorized as a spatiotemporal sequence forecasting problem. Given a cloud map of a past period, the spatial features and temporal dynamics of the current period are learned, and then, the cloud map sequence of the same region is predicted for a future period, demonstrating that RingMo-Sense has the same excellent capability in the field of cloud map forecasting.

1) Specific Architecture: For the cloud forecasting task, we use exactly the same model architecture as video prediction, as shown in Fig. 6. The model structure used for cloud prediction is also an adaptation of the classical model SimVP [63] used for natural scenes, whose specific parameter settings are described in Section V-C3.

2) Dataset Introduction: We adopt a satellite-based dataset, which is presented by Nielsen et al. [44] and designed for cloud forecasting. The CloudCast dataset provides 70 080 images containing 11 different cloud types. Its resolution is 928 × 1530 pixels (each pixel represents a 3 × 3 km^2 area), recorded at 15-min intervals over the 2017–2018 period. This work also includes a downsampled low-resolution dataset of 128 × 128 (15 × 15 km^2) to support small-scale experiments and analyses. In this article, we experiment with the small version.

TABLE III
COMPARISONS WITH STATE-OF-THE-ART METHODS ON CLOUD FORECASTING

Model	SSIM↑	PSNR↑
PredRNN [66]	0.9633	41.23
RingMo-Sense(-J)	0.9615	41.05
RingMo-Sense(-TO)	0.9584	40.89
RingMo-Sense(-SO)	0.9363	39.99
RingMo-Sense(-ST)	0.9362	39.99
RingMo-Sense(-STT)	0.9629	41.16
RingMo-Sense	0.9657	41.52

3) Implementation Details: In our experiments, we take 32 frames from the sequential cloud images as one sample, the first 16 as input and the rest frames as labels, with a size of 128 × 128. We used the Adam optimizer by default and the OneCycleLR learning rate strategy with a default learning rate of 0.01 and a batch size of 4, and trained 51 epochs. We use SSIM and peak signal-to-noise ratio (PSNR) to evaluate the forecasting performance.

4) Comparisons With SOTA Methods: Table III shows the accuracy of our model compared to other models on the cloud forecasting task. PredRNN proposes an ST-LSTM and allows cross-layer interaction of the in-memory states of different LSTMs, and it achieves an SSIM of 0.9633 on the Cloudcast dataset but not as good as when loaded with the RingMo-Sense pretraining parameters, which can achieve 0.9657.

5) Ablation Studies: Table III shows the results of the ablation experiments for the cloud prediction task. The PSNR for the RingMo-Sense configuration is 41.52, which is 0.47 higher than that of RingMo-Sense(-J), proving the effectiveness of our proposed progressive training paradigm; it is 0.63, 1.53, and 1.53 higher than that of RingMo-Sense(-TO), RingMo-Sense(-SO), and RingMo-Sense(-ST), respectively, proving the effectiveness of the spatial branch and the temporal branch and the spatiotemporal branch, respectively. The PSNR of the RingMo-Sense(-STT) configuration is 41.16, which is 0.36 lower than that of RingMo-Sense, proving that our proposed spatiotemporal mask strategy is better than the tubewise strategy for extracting spatiotemporal features.

6) Visualization Results: The visualization results of PredRNN and RingMo-Sense in the cloud prediction task are shown in Fig. 9. It is obvious that RingMo-Sense's prediction results are closer to the true values, and the images are clearer. This also approves again the powerful extraction capability of RingMo-Sense's pretraining parameters for spatiotemporal features. Fig. 10 shows the mean value of the predicted results per frame for a random sample in the CloudCast dataset using our method and the PredRNN, which shows that our model's predicted results are much closer to the ground truth. However, our model also suffers from the shortcoming that the predictive power of the model diminishes as the prediction

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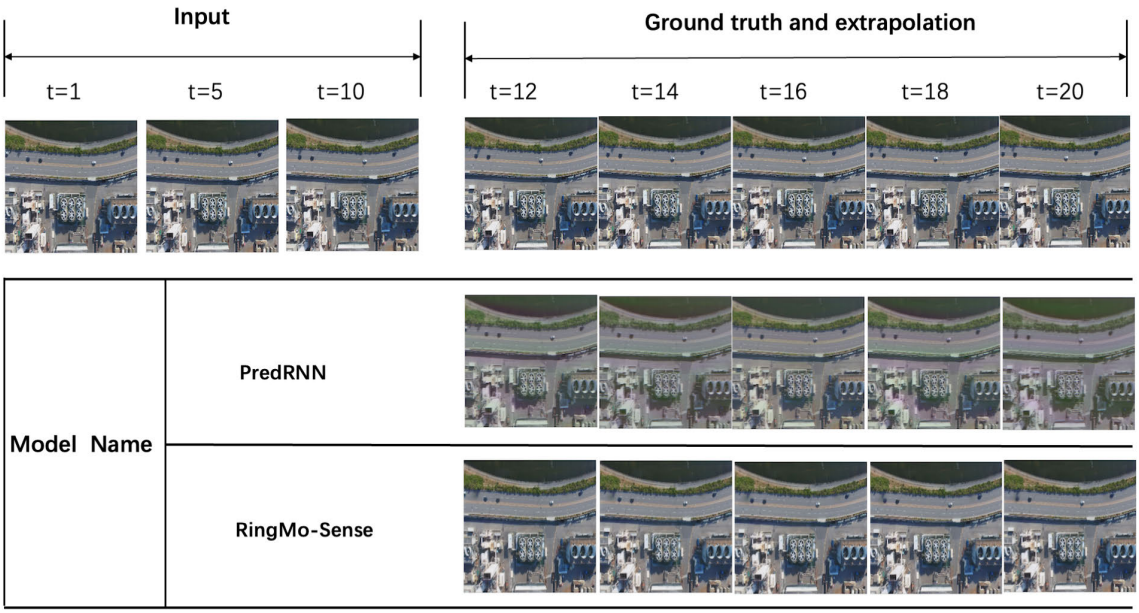


Fig. 7. Visualization of the prediction results on the MOR-UAV dataset. The first three figures on row 1 are the input frames at time step 1, time step 5, and time step 10, respectively; the rest in line 1 are the ground-truth frames at five different time steps. The last two rows show the prediction results of PredRNN and RingMo-Sense.

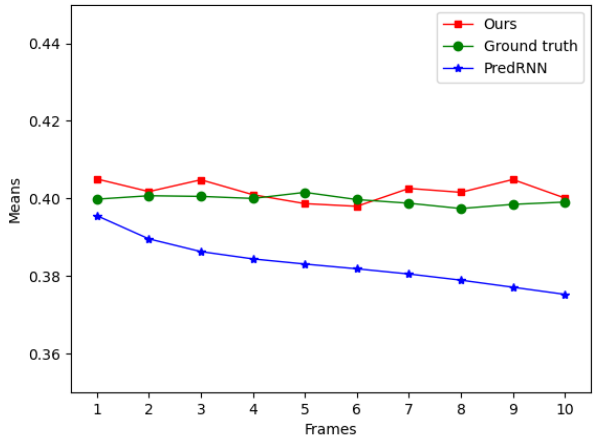


Fig. 8. Visualization of the mean value of prediction results on the MOR-UAV dataset. The green line represents the mean value of ground-truth frames, the red line represents the mean value of our model's prediction, and the blue one represents that of PredRNN.

time progresses. In the future, we will address this issue by increasing the loss weights of the leaning frames.

D. Radar Echo Extrapolation

Flooding and geological hazards caused by heavy convective weather (short-term heavy precipitation, thunderstorms, tornadoes, and hailstorms) are a great danger to people's lives and property, and thereby, it is of great significance to provide proximity forecasts and warnings for them. Radar extrapolation forecasting has become a key technical support [67], [68], [69], [70], [71] for short-time quantitative precipitation proximity forecasting. It refers to the determination of the movement of echoes (e.g., storm cells and precipitation areas) based on the echo data detected by weather radar and to forecast the state of radar echoes after a certain period of time by linear or nonlinear extrapolation of the echoes.

1) Specific Architecture: For the radar echo extrapolation task, we use exactly the same model architecture as video prediction, as shown in Fig. 6. The model structure used for radar echo extrapolation is also an adaptation of the classical model SimVP [63] used for natural scenes, whose specific parameter settings are described in V-D3.

2) Dataset Introduction: We used the HuaBei2021 dataset to fine-tune the foundation model for the radar echo extrapolation task. The raw data are nationwide radar reflectivity images from June to August 2021 (rainy season), which covers an area of 109.9°–123.0° East longitude and 32.98°–46.14° North latitude. However, such a map is distorted on the actual surface curves, which leads to the tedious task of recalculating the formula for surface spreading each time a small area is boxed if the prediction is made directly on the plane. Thus, we process this raw image into a grid of data divided by latitude and longitude, which made subsequent operations very easy. Due to the large national scale area, we selected 256 $\times$ 256 data covering northern China as our experimental data and later selected samples with radar echo intensity greater than a certain threshold for selection, resulting in the HuaBei2021 dataset.

3) Implementation Details: For the experiments, we selected 20 consecutive radar echo maps as a training sample, where the first ten are the inputs to the model and the last ten are the predicted targets, i.e., the labels corresponding to the inputs. The loss is the most parsimonious MSE, and the learning rate is 0.001. The learning rate decay scheme is OneCycleLR so that the learning rate varies between set extremes rather than decreasing monotonically to prevent the model from falling into a local optimum solution. The batch size is set to 4, and the number of training epochs is set to 51. In addition to the well-known and widely used

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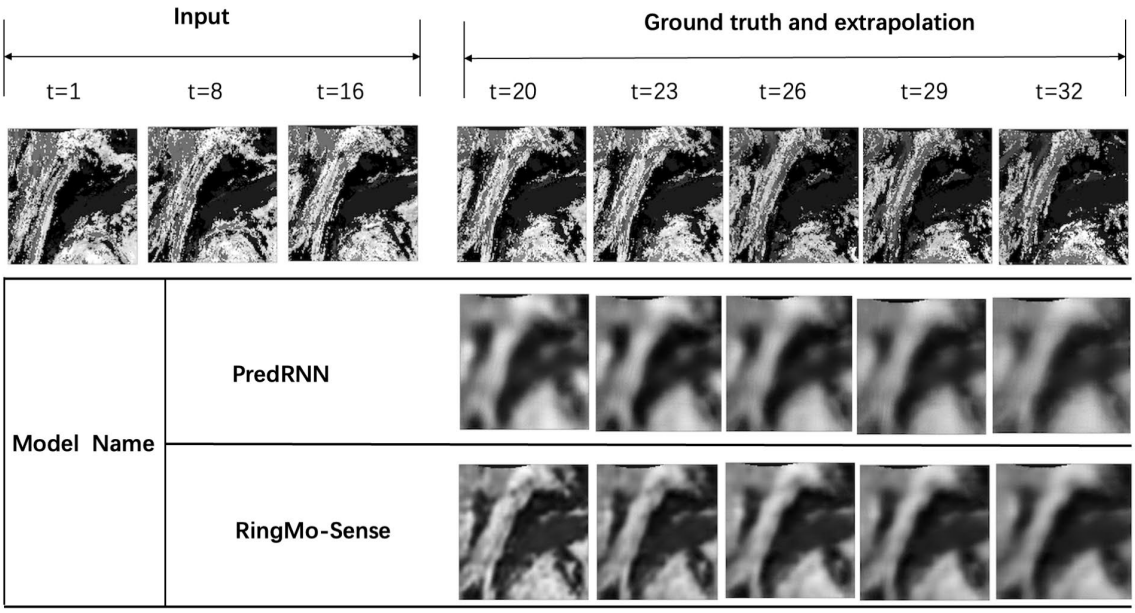


Fig. 9. Visualization of the forecasting results on the Cloudcast dataset. The first three pictures on row 1 are the input frames at time step 1, time step 8, and time step 16, respectively; the rest in line 1 are the ground-truth cloud maps at five different time steps. The last two rows show the forecasting results of PredRNN and RingMo-Sense.

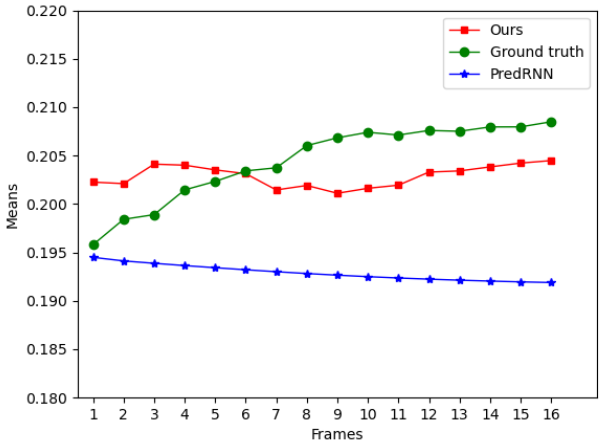


Fig. 10. Visualization of the mean value of prediction results on CloudCast. The green line represents the mean value of ground-truth frames, the red line represents the mean value of our model's prediction, and the blue one represents that of PredRNN.

TABLE IV
CONTACT NUMBERS

		Alarms	
		Yes	No
Events	Yes	X	Y
	No	Z	W

MSE and the SSIM mentioned above, we also utilized three other commonly used evaluation metrics in the field of meteorology, i.e., probability of detection (POD), the false alarm rate (FAR), and critical success index (CSI).

As shown in Table IV, X represents times an event has occurred for which a warning has been issued in advance. Y represents the times that the event occurred, but the violator gives a warning. Z represents the number of times that an event did not happen, but a warning was issued by mistake. W represents the number of times that no warning was issued and the event that did not happen

$$\text{POD} = \frac{X}{X+Y}. \tag{4}$$

The best possible value for POD is 1, the worst possible value is 0, and the FAR represents the proportion of the warnings given that the event did not happen

$$\text{FAR} = \frac{Z}{X+Z}. \tag{5}$$

The worst possible value of FAR is 1. The fewer the false alarms, the smaller the FAR, and the better the outcome, with the best possible value being 0. Too many alarms will result in a high POD value but a high FAR value at the same time. Too few alarms will result in a small FAR value but a small POD value. The combined effect of POD and FAR is represented by a parameter called the success index CSI (also commonly known as the threatscore (TS) score)

$$\text{CSI} = \frac{X}{X+Y+Z}. \tag{6}$$

4) Comparisons With SOTA Methods: Table V shows the accuracy of our model in the radar echo extrapolation task compared to other models. ConvLSTM achieves an MSE of 57.5659 on the HuaBei2021 dataset. In our model, when loaded with RingMo-Sense pretraining parameters, MSE is significantly reduced to 52.8029, which demonstrates the power of RingMo-Sense in extracting spatiotemporal features effectively in radar echo tasks.

5) Ablation Studies: Table V shows the results of the ablation experiments for the radar echo extrapolation task. POD for the RingMo-Sense configuration is 0.027 higher

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TABLE V
COMPARISONS WITH STATE-OF-THE-ART METHODS ON RADAR ECHO EXTRAPOLATION

Model	MSE↓	SSIM↑	POD↑	FAR↓	CSI↑
ConvLSTM [72]	57.5659	0.9455	0.2101	0.3773	0.1900
RingMo-Sense(-J)	54.3722	0.9485	0.2280	0.3677	0.2174
RingMo-Sense(-TO)	59.5840	0.9385	0.1449	0.4102	0.1286
RingMo-Sense(-SO)	58.4899	0.9405	0.1705	0.4116	0.1494
RingMo-Sense(-ST)	53.5347	0.9464	0.2521	0.3474	0.2216
RingMo-Sense(-STT)	53.3117	0.9468	0.2542	0.3455	0.2243
RingMo-Sense	52.8029	0.9480	0.2550	0.3591	0.2246

than that of RingMo-Sense(-J), which proves the effectiveness of our proposed recursive training paradigm, 0.1101, 0.0845, and 0.0029 higher than that of RingMo-Sense(-TO), RingMo-Sense(-SO), and RingMo-Sense(-ST), respectively, which proves the effectiveness of the spatial branch, the temporal branch, and the spatiotemporal branch, respectively. The POD of the RingMo-Sense(-STT) configuration is 0.0008 lower than that of RingMo-Sense, which proves that our proposed spatiotemporal mask strategy is better than the tubewise strategy for extracting spatiotemporal features.

6) Visualization: Fig. 11 shows the visualization results of the radar echo extrapolation task. The radar echo evolves over time, which requires the model to not only extract its spatial pattern but also capture its dynamic pattern. The prediction results output by ConvLSTM in the figure are not only fuzzy but also differ from the true value, while RingMo-Sense gives a clearer and more accurate result. Fig. 12 shows the mean value of the predicted results per frame for a random sample in the HuaBei2021 dataset using our method and the PredRNN, which shows that our model's predicted results are much closer to the ground truth. The shortcoming is that the echo maps predicted by our model are still blurred, and more sharpened features may be generated in the future by adding image spatial gradient loss.

E. Object Detection in Satellite Videos

The object detection task has developed rapidly with the development of convolution neural networks [73], [74], [75], [76], [77]. Practical application requirements have gradually developed from image object detection to video object detection [51], [78], [79], [80], [81], [82]. If we directly apply the still image detector to each video frame, the detection accuracy will be hindered because the video stream changes over time. Therefore, the important temporal information of the video itself needs to be used to compensate for further optimization. In this section, to evaluate the performance of the foundation model and verify its effectiveness for the video object detection task, the necessary detection heads are assembled on the foundation model to develop an object detector suitable for satellite videos.

1) Specific Architecture: As shown in Fig. 13, the architecture of our video object detector is built on the YOLOX [83] framework, which is widely studied and applied in real-world scenarios. First, the two reference frames F_{R1} and F_{R2} and the target frame F_T are input into the backbone network and generate deep feature representations. The 3-D Swin Transformer consists of four cascaded stages. Then, the features $[P2, P3, P4]$ produced by the last three stages are sent to the FPN to gain more powerful representations. Next, the features carrying rich spatial–temporal contexts are fed into the detection heads. Finally, the detection bounding boxes on the target frame can be obtained.

TABLE VI
COMPARISONS WITH DIFFERENT METHODS ON SATELLITE VIDEO OBJECT DETECTION. AP_{50} (%) IS CALCULATED ON HORIZONTAL BOUNDING BOXES

Detector	AP_{50}(%)	AP^{S}_{50}(%)	AP^{M}_{50}(%)
Faster R-CNN [73]	59.0	53.4	73.5
Cascade R-CNN [86]	60.9	49.4	78.0
RetinaNet [87]	60.6	47.2	81.7
DSFNet [88]	61.3	57.0	72.3
YOLOX [84]	62.8	37.0	81.3
RingMo-Sense(-J)	64.7	40.1	80.6
RingMo-Sense(-TO)	62.3	31.9	79.0
RingMo-Sense(-SO)	63.6	38.5	78.6
RingMo-Sense(-ST)	64.3	38.3	79.1
RingMo-Sense(-STT)	62.5	40.2	78.0
RingMo-Sense	65.9	41.3	80.8

2) Dataset Introduction: AIR-MOT [84] is a challenging video dataset covering many cities in the world, and the videos are obtained by the Jilin-1 satellite. We filter out the videos in the AIR-MOT dataset, which are almost empty or contain little information. As a result, a total of 81 videos are retained, and the total number of instances including two categories (i.e., the aircraft and ship) is 1337. Each video has 70–326 frames with a size of 1920 × 1080 pixels. The original annotations are prepared for the task of multiobject tracking in satellite videos. Thus, we extract the location and category information of each instance and convert them into the annotation format required for the video object detection task. Besides, we randomly select 69 videos to train the model, and the left 12 videos are the test set.

3) Implementation Details: In our experiments, each input video frame is resized into 1024 × 640 pixels. We train our model using multiscale images, and the shortest side ranges from 576 to 960 pixels. The adopted optimizer is SGD, the weight decay is rendered to 5×10^{-4}, and the momentum is set to 0.9. The training process is with a learning rate of 0.002 and a minibatch of 4. The training schedule is 20 epochs, and the total time of the training phase is about 70 h. We evaluate the detection performance by the commonly used metric average precision (AP). AP^{S}_{50} and AP^{M}_{50} are for small and medium objects. In addition, small objects have an area smaller than 32 × 32 pixels, and medium objects have an area between 32 × 32 pixels and 96 × 96 pixels.

4) Comparisons With SOTA Methods: To demonstrate the effectiveness of the video detector, we perform many comparisons with other widely used methods, and the results are reported in Table VI. When we utilize the baseline detector YOLOX [83] to perform detection on each video frame, the AP_{50} can achieve 62.8%, and its accuracy is better than other two- or one-stage image detectors, which also shows the strong

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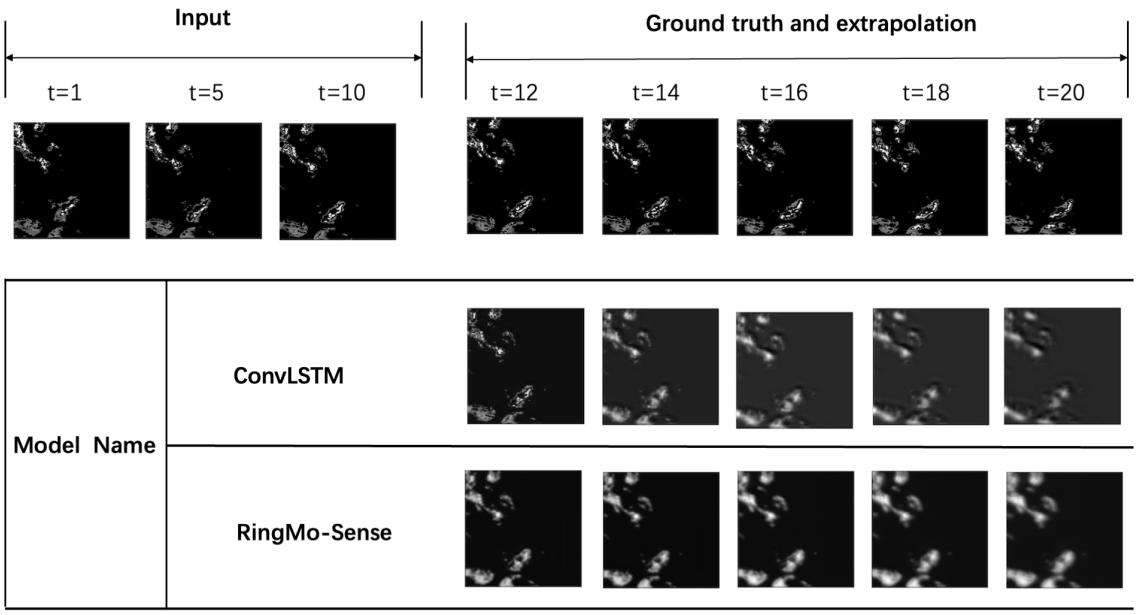


Fig. 11. Visualization of the forecasting results on the HuaBei2021 dataset. The first three pictures on row 1 are the input echo maps at time step 1, time step 5, and time step 10, respectively; the rest in line 1 are the ground-truth echo maps at five different time steps. The last two rows show the forecasting results of ConvLSTM and RingMo-Sense.

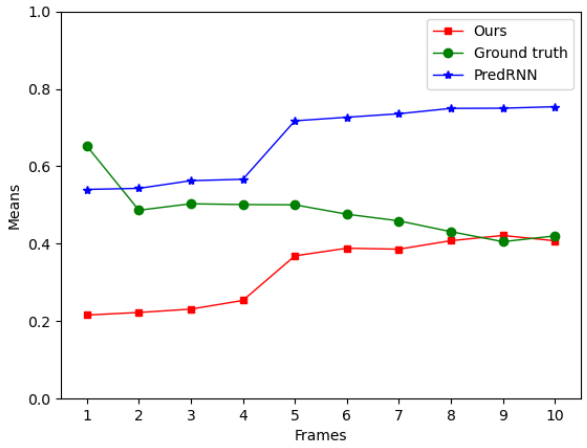


Fig. 12. Visualization of the mean value of prediction results on the HuaBei2021 dataset. The green line represents the mean value of ground-truth frames, the red line represents the mean value of our model's prediction, and the blue one represents that of PredRNN.

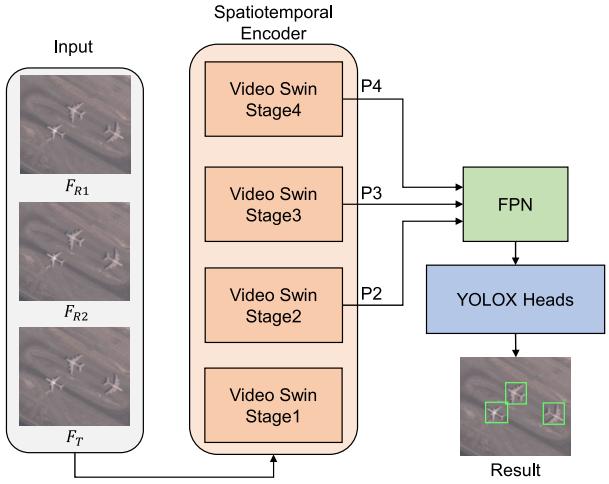


Fig. 13. Architecture of the video object detector with the foundation model. The detector is built on YOLOX, adopting the encoder of the foundation model as the backbone. F_{R1} and F_{R2} represent two reference frames, and F_T denotes the target frame.

spatial feature extraction ability of YOLOX and the reason why it is popular in the current object detection field. Due to the introduction of temporal semantic descriptions brought by the foundation model, the detector YOLOX with the pretrained backbone RingMo-Sense can yield a better AP_{50} of 65.9%.

5) Ablation Studies: In this part, we provide some necessary ablation studies to investigate the effect of the pretrained model using different strategies on the detection task. The ablation results are also listed in Table VI. Specifically, AP_{50} of the detector with RingMo-Sense(-J) drops 1.2% because RingMo-Sense(-J) adopts a three-stream joint training strategy instead of the progressive training paradigm. With the pretrained models using different streams, the detector with RingMo-Sense(-TO), RingMo-Sense(-SO), and RingMo-Sense(-ST) can reach 62.3%, 63.6%, and 64.3% in terms of AP_{50}, respectively. It can be observed that they are less than 1.6% compared to the RingMo-Sense, which indicates that the progressive pretraining of the three streams is necessary to achieve the promising performance. Besides, the detector with the RingMo-Sense(-STT) stays at an AP_{50} of 62.5%, with a reduction of 3.4%, demonstrating that the spatiotemporal joint mask strategy can help capture complementary semantics and further improve the generalization ability of the model.

6) Visualization Results: Some detection results are visualized in Fig. 14. The image detector only focuses on modeling spatial information inside a static image, resulting in the false positives (FPs) of some similar facilities. Compared to it, the method with the pretrained RingMo-Sense can produce more accurate boxes due to a better understanding of objects through the joint learning of temporal and spatial information.

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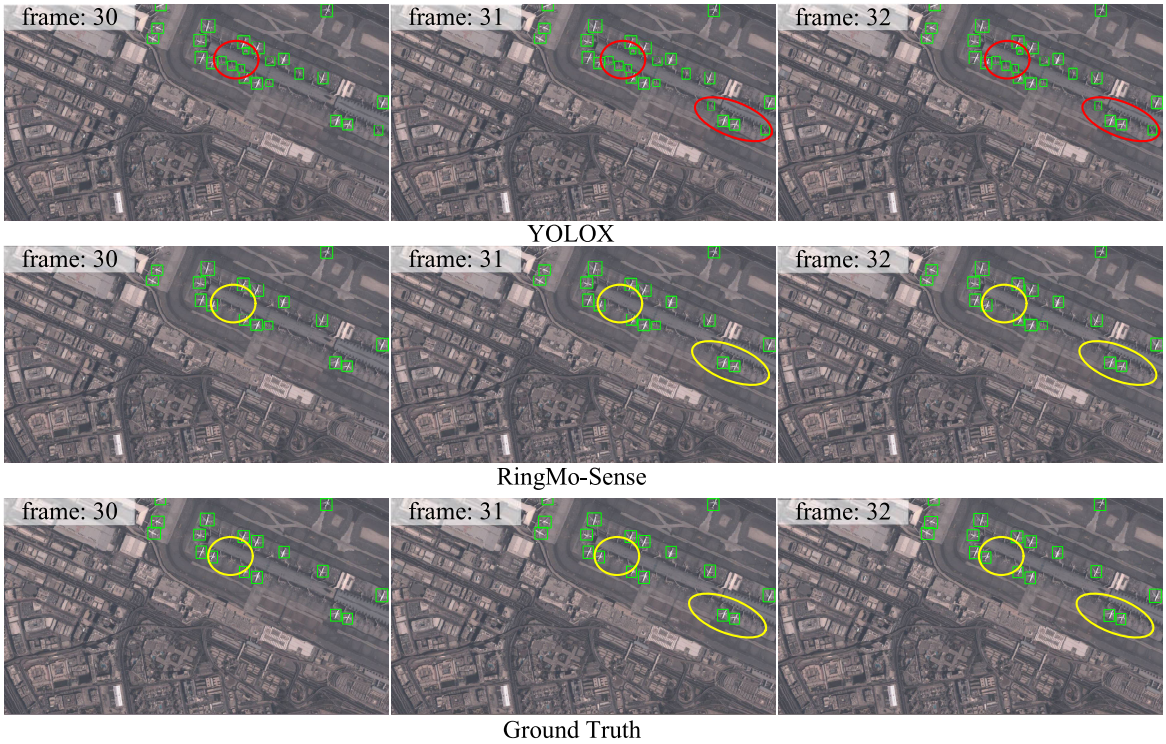


Fig. 14. Visualization of detection results on the AIR-MOT dataset. The three sequences contained in the first row are the visualization produced by the detector YOLOX. The three sequences contained in the second row are the visualization of the method with the foundation model RingMo-Sense.

However, there are still several missed detections and false boxes, and we intend to tackle these challenges by further mining instance-level semantic information of consecutive frames in the future.

F. Multiobject Tracking in Satellite Videos

The main purpose of the multiobject tracking task [88], [89], [90] is to detect the potential targets and associate them with identities across video frames. Recently, with the significant progress made in aerospace information technology, high-resolution satellite videos have become available, which also provides opportunities for ground object tracking tasks in large-scale scenes. Among the modern trackers, the pipeline that first performs detection on each video frame and then postprocesses them into tracking results gradually comes to the fore. Accordingly, we utilize the existing association methods on the detection boxes generated in Section V-E to obtain the tracking results. Then, we can further explore the effect of the foundation model on the multiobject tracking task, in which the spatiotemporal modeling capability plays an important role.

1) Specific Architecture: We employ a simple and effective association algorithm for tracking, i.e., BYTE [91], which boosts the tracking performance by taking every detection box into account instead of partial ones. Without any tricks and modifications in the decoder part of the downstream task, our experimental results are more indicative of the contribution of our foundation model.

2) Dataset Introduction: As described in Section V-E2, the detection results are achieved on AIR-MOT [84]; after performing the association algorithm, we can obtain the values of corresponding tracking indicators using the original tracking annotations, whose format is similar to MOT benchmark [92]. The total number of trajectories for 81 videos is 1337, and the division of the dataset has been elaborated.

TABLE VII
COMPARISONS WITH DIFFERENT METHODS ON MULTIOBJECT TRACKING TASK

Tracker	MOTA↑	IDF1↑	FP↓	FN↓	IDs↓
SORT [94]	47.8%	67.8%	1756	9868	7
DeepSORT [95]	48.0%	68.1%	1806	9780	2
MOTDT [96]	48.6%	68.4%	1562	9871	2
BYTE [92]	49.8%	70.2%	2143	9040	2
RingMo-Sense(-J)	52.7%	69.4%	159	10363	0
RingMo-Sense(-TO)	52.5%	69.8%	728	9845	9
RingMo-Sense(-SO)	51.9%	68.4%	273	10432	7
RingMo-Sense(-ST)	51.7%	68.8%	652	10089	11
RingMo-Sense(-STT)	51.4%	68.5%	591	10200	20
RingMo-Sense	53.2%	70.3%	466	9955	0

3) Implementation Details: In our experiments, the objects with a detection score greater than 0.5 are accepted, and we render the matching threshold for tracking to 0.8. The lost tracks are kept for 30 frames to cope with their reappearance. Besides, tiny detection boxes whose area is less than 50 pixels are filtered out. To compare the performance of various trackers, we utilize several quantitative metrics, including multiple object tracking accuracy (MOTA), ID F1 score (IDF1), the number of FPs, false negatives (FNs), and IDentity switches (IDs).

4) Comparisons With SOTA Methods: Table VII shows the tracking accuracy of our method and other different trackers. Through associating low-score detection boxes with tracklets, BYTE [91] yields better accuracy (i.e., 49.8% on MOTA

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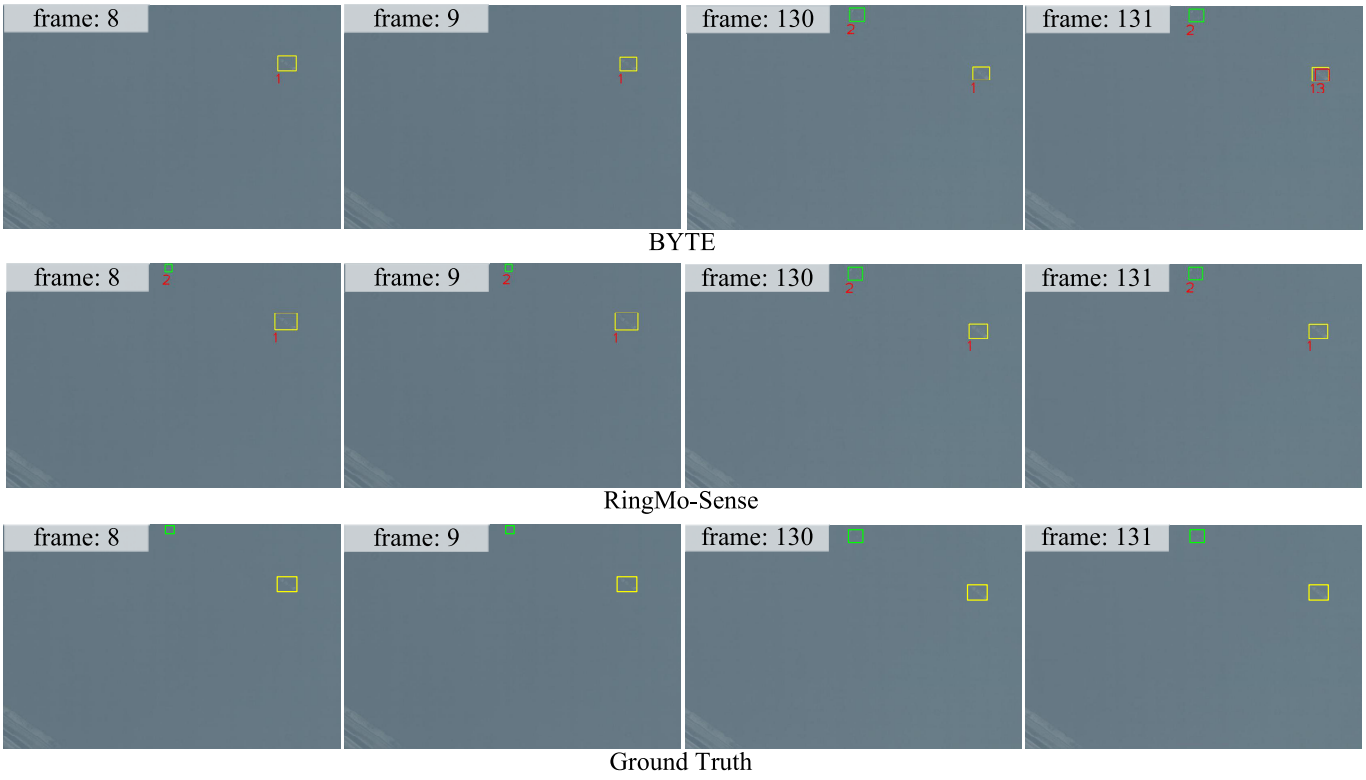


Fig. 15. Visualization of tracking results on the AIR-MOT dataset. The four sequences contained in the first row are the visualization of the baseline BYTE. The four sequences contained in the second row are the visualization of the method with the foundation model RingMo-Sense.

and 70.2% on IDF1) than other tracking methods, including SORT [93], DeepSORT [94], and MOTDT [95]. Equipped with pretrained weights of RingMo-Sense, the accuracy of the baseline BYTE is further boosted. It lifts the corresponding indicators, i.e., from 49.8% to 53.2% on MOTA and from 70.2% to 70.3% on IDF1. The experimental results show that the pretrained RingMo-Sense can facilitate tracking methods after encoding rich temporal and spatial information.

5) Ablation Studies: To investigate the effects of different pretrained weights on the multiobject tracking task, we also conduct ablation studies of RingMo-Sense, and the results are shown in Table VII. Because the temporal and spatial information encoding is difficult without the progressive training paradigm, the tracking results with RingMo-Sense(-J) only yield an MOTA of 52.7% and the IDF1 of 69.4%. We can find that the model with RingMo-Sense(-ST) decreases MTOA by 1.5%, indicating that the spatiotemporal stream is necessary to bring desirable performance improvement. In addition, the tracking results with RingMo-Sense(-STT) drop 1.8% on IDF1, demonstrating that the cubewise masking method works, not just adding a training stream.

6) Visualization Results: Some visualization results are reported in Fig. 15 to further demonstrate the gains that RingMo-Sense brings to the tracking task. Due to weather reasons such as heavy fog, the video imaging results are affected. This leads to a difficult case both for detection and tracking tasks. In frames 8 and 9 of the test video, the ship-2 is blurry, and it is lost by the baseline BYTE [91]. After loading the pretrained RingMo-Sense, the ship-2 can be tracked stably due to robust spatial semantics. In frame 131, the baseline yields an FP (i.e., ship-3). The model with RingMo-Sense can alleviate it because of its spatiotemporal modeling capabilities.

G. Remote Sensing Time-Series Images' Segmentation

A large part of aerial space information extraction is to use RS platforms to obtain earth observation data and extract surface information from the data of satellite-based or airborne platforms. In our aerial space data proposition, the information on landscape distribution and change is important supporting information for making ecological or habitat decisions such as natural resources and urban management. In the past, it took a lot of human and material resources to collect information on the distribution and change of land features, and the timeliness was not high. Based on the current rapidly developing deep learning methods, the semantic segmentation-based method for land feature information has become the mainstream means at present. Due to the current development of the means to obtain the time-series data, the amount of relevant data acquisition is increasing day by day, and semantic segmentation based on time-series data is gradually becoming a new hot issue. In RS, there are relatively few relevant datasets and methods available. As far as we are aware, the temporal segmentation tasks in the UAV scenario are primarily conducted using the UAVid dataset, which is why we have also selected it as our benchmark dataset.

1) Specific Architecture: We adopt a simple and effective temporal semantic segmentation model in computer vision as the framework, i.e., the CFFM [96] model. The model follows the typical encoder–decoder pipeline for image semantic

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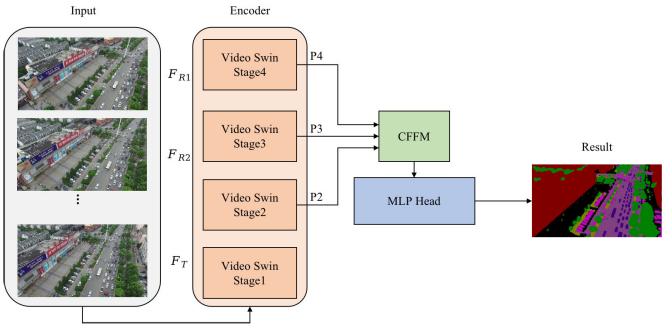


Fig. 16. Architecture of the CFFM, adopting the encoder of the foundation model as the backbone. I_t is current image to be segmented; I_{t-k_1}, I_{t-k_2}, and I_{t-k_3} represent three reference frames; and F_t denotes the final feature map of I_t.

segmentation, utilizing the visual encoder (also known as the backbone) to extract the intermediate feature map from each frame. Afterward, a straightforward MLP decoder is employed to segment the current frame. In addition, considering the spatiotemporal information of the temporal data, the method adds the cross-frame feature mining (CFFM) module to fully exploit the temporal features. However, the intermediate feature maps extracted by the encoder are independent. We believe that the temporal interframe interaction in the encoder part can make fuller use of the temporal correlation, which can guide the extraction of each frame feature and obtain better image features based on temporal correlation. The temporal-guided image features can better adapt to the characteristics of the temporal image segmentation task and obtain better segmentation performance, which is demonstrated by our experiments.

In this study, we compare CFFM-RingMo-Sense (Fig. 16), which utilizes the RingMo-Sense foundation model in its segmentation framework with the CFFM model. Specifically, we modify only the encoder portion of the model while maintaining consistency with the original method in the decoder module. Despite the modification, the number of parameters remains largely unchanged from the original model. Notably, our experimental results reveal that the enhanced performance of our model can be attributed to the robustness and effectiveness of our pretrained model, which is achieved without any additional tricks or modifications.

The best-performing model on the UAVid dataset [5] is UNetFormer [97]; since it relies on a single image only and uses a different paradigm, we only use it as a baseline for comparison.

2) Dataset Introduction: The UAVid dataset [5] is a high-resolution oblique view town scene video dataset acquired based on UAVs. The shooting pitch angle of the dataset is set to about 45, and the flight height is kept at 50 m or less as much as possible. The resolution of the dataset is 4096 × 2160 or 3840 × 2160, and it contains eight types of targets: building, road, tree, low vegetation, static car, moving car, human, and clutter. Since the video segmentation annotation is time-consuming and laborious, the annotation of the UAVid dataset is not continuous but every 100 frames. Although the remaining frames are not annotated, they are provided in the form of video data for researchers who need them. The dataset consists of a total of 42 video sequences, which are divided into three sets: 20 for training, seven for validation, and 15 for testing. All annotations included in the dataset adhere to the standard criteria typically used in the field of video semantic segmentation.

It is worth mentioning that, due to the high resolution of the dataset and the resource-intensive nature of training, the training set was cropped in accordance with the pipeline outlined in [96].

3) Implementation Details: In training settings, the data input was first scaled to 1024 × 2048 and then cropped to a resolution of 480 × 480. The training process involved feeding four frames of images at a time. Data augmentation techniques similar to those used in [96] were applied, and the AdamW optimizer was used with a learning rate of 0.00006. The total batchsize was set to 8, and the learning rate was decayed using a poly strategy with a power of 1. In addition, the encoder component was initialized with pretrained model parameters from the pretrain section. In test settings, the dataset input size is the original size, and no test enhancements are used.

Evaluation Metrics mIoU: The evaluation metric employed in this study is the widely-used mean Intersection over Union (mIoU) metric, which is also the accuracy metric used in the UAVid dataset. The mIoU is calculated by taking the Intersection over Union (IoU) of the predicted segmentation mask and the ground-truth mask for each class and then averaging the IoUs across all classes. mIoU is calculated as follows:

$$\text{mIoU} = \frac{1}{k+1} \sum_{i=0}^{k} \frac{p_{ii}}{\sum_{j=0}^{k} p_{ij} + \sum_{j=0}^{k} p_{ji} - p_{ii}} \tag{7}$$

where k+1 is the number of classes and p_{ij} means that the probability of predicting class i to class j.

4) Comparisons With SOTA Methods: To showcase the effectiveness of our pretrained hit model, we conducted several experiments. Specifically, we compared the performance of the CFFM model with pretraining weights to that of other models, and the corresponding results are presented in Table VIII.

The temporal semantic segmentation framework in computer vision is applied to solve the temporal segmentation problem in RS scenarios, achieving comparable performance in mIoU metrics. Our model, compared with UNetFormer on this dataset, leverages spatiotemporal information to assist image segmentation. By introducing spatiotemporal association guidance at an earlier stage (encoding stage), our

TABLE VIII
COMPARISONS WITH COMPARABLE METHODS ON TIME-SERIES IMAGE'S SEGMENTATION

Method	mIoU(%) ↑	mDice (%) ↑
UNetformer[99]	70.0	-
CFFM[97]	68.35	78.98
RingMo-Sense(-J)	69.07	80.41
RingMo-Sense(-TO)	68.33	79.74
RingMo-Sense(-SO)	68.66	80.01
RingMo-Sense(-ST)	69.02	80.43
RingMo-Sense(-STT)	68.61	79.94
RingMo-Sense	69.16	80.19

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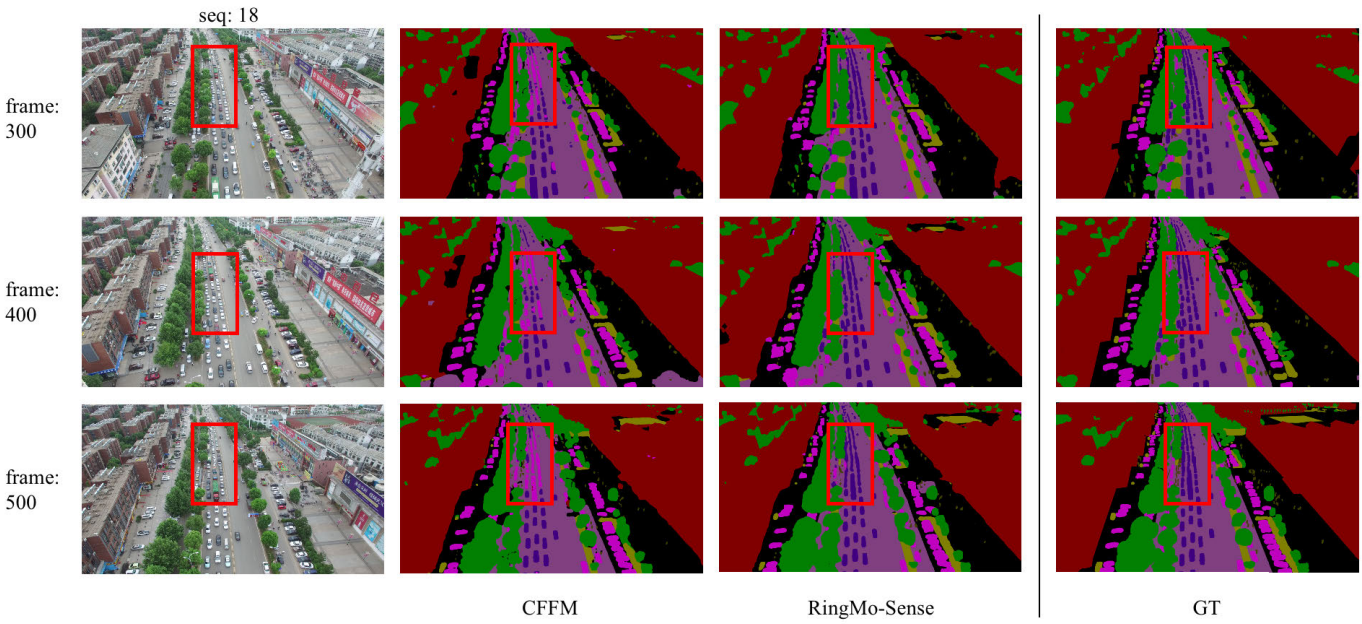


Fig. 17. Visualization of segmentation results on the UAVid dataset. The first column contains some images of Seq18, followed by the segmentation results of each model, respectively.

model better extracts spatiotemporal-related image features, and the experimental results demonstrate that this can effectively improve model performance. While the backbone of the original CFFM model is already robust, our model improves training time and reduces memory consumption without hyperparameter adjustment.

5) Ablations Studies: We compare the performance of CFFM-RingMo-Sense with pretraining weights obtained from different datasets and different pretraining strategies. The results are presented in Table VIII. To investigate the impact of weights obtained from different training methods on the temporal segmentation task, we conduct ablation experiments using several different sets of training strategies.

Based on the comparison results presented in Table VIII, we can infer that the effect of different pretraining strategies on the Time-Series Images' Segmentation task is not significant. This may be attributed to the fact that our dataset is relatively easy and not too complex, which allows our model to quickly learn the semantic knowledge without requiring additional pretraining information from larger datasets. Specifically, the pretraining weights obtained from the RingMo-Sense training scheme show the best performance, and the RingMo-Sense(-ST) pretraining weight, trained on our dataset, performs similar to RingMo-Sense. However, we observed a significant decrease in performance for RingMo-Sense(-TO) without spatial branching, indicating that spatial information is relatively more critical for the Time-Series Images' Segmentation task. Therefore, we can conclude that the scenario-specific dataset can improve the downstream task's performance, and RingMo-Sense's weights capture more temporal information in the dataset, thanks to the special spatiotemporal mask trick. In Time-Series Images' Segmentation, the spatial information of the current frame is more crucial, and the spatiotemporal information is only supportive. Therefore, spatiotemporal features learned based on various forms show similar performance.

6) Visualization Results: To demonstrate the effectiveness of our pretrained weights in time-series segmentation, we provide visualizations of the segmentation results in Fig 17. The figure shows the original image at the top, with the model's segmentation results and ground truth displayed in the columns below. The comparison reveals that our pretrained model significantly outperforms the baseline CFFM model in segmenting the red boxed region, producing more finely segmented results. This region contains a moving car, and our model's ability to accurately capture this dynamic target class suggests that it has learned better temporal information. Overall, the visualizations confirm that our pretrained weights are indeed beneficial for time-series segmentation. However, the segmentation edges generated by our model exhibit jaggedness and blurriness, which may be due to the relatively small spatial resolution size used during training. This could have resulted in the model learning spatial features that are insufficiently rich and accurate.

VI. Conclusion

In this article, we propose RingMo-Sense, a foundation model for spatiotemporal prediction in the RS domain. RingMo-Sense utilizes spatial, temporal, and spatiotemporal triple-branch structures for spatiotemporal evolution disentangling and realizes parameter-sharing of multiple branches by specific masking and progressive joint training strategies. In addition, we construct an RS spatiotemporal dataset, which serves for pretraining of RingMo-Sense. After self-supervised pretraining, the RingMo-Sense is fine-tuned and achieves competitive performance in several downstream tasks, demonstrating the effectiveness of the model. In future work, we will continue to improve the representation capabilities of the RS foundation model for spatiotemporal prediction and work on extending the RingMo-Sense to more downstream applications to obtain a more pervasive performance for spatiotemporal prediction in RS.

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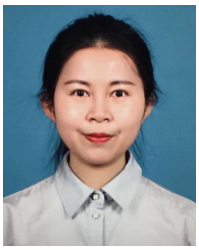
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